

A video container usually contains multiple tracks — a video track (without audio), plus one or more audio tracks (without video). These tracks are interrelated. An audio track contains markers within it to help synchronize the audio with the video. Individual tracks can have metadata, such as the aspect ratio of a video track, or the language of an audio track. Containers can also have metadata, such as the title of the video itself, cover art for the video, episode numbers (for television shows), and so on.



Overview of a Video Container

The figure below shows how a video file (a video container) contains audio tracks, video tracks, and additional metadata.

Some of the popular video container formats include the following:

- ▶ MPEG 4 (.mp4 or .m4v) - The MPEG 4 container is based on Apple's older QuickTime container (.mov).
- ▶ Flash Video (.flv) - This is the most common of all and is used by Adobe Flash. Earlier this was the only container format that Flash supported. More recent versions of Flash also support the MPEG 4 (.mp4) container.
- ▶ Ogg (.ogv) - It is an open standard, open source-friendly, and unencumbered by any known patents. The Ogg video is called "Theora" while the Ogg audio is called "Vorbis".
- ▶ WebM - This is a new container format announced in May, 2010. It is technically similar to another format, called Matroska (.mkv). It has been designed to be used exclusively with the VP8 video codec and Vorbis audio codec.
- ▶ Audio Video Interleave (.avi) - The AVI container format was invented by Microsoft in a simpler time, when the fact that computers could play video at all was an amazing thing.

Audio and Video Codecs

When you are "watching a video," your video player is doing at least three things at once:

1. Interpreting the container format to find out which video and audio tracks are available, and how they are stored within the file so that it can find the data it needs to decode next